

PROJECT COMPETITION 2026

Department of Information Technology | Techno Main Salt Lake

- **Problem Statement Title** – Develop a system to securely encrypt and decrypt files cryptographic algorithms.
- **Team Number** - 41
- **Team Name** – Secure Minds
- **Student Name – University Roll with Year of Registration**
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Secure File Encryption and Decryption System (CRYPTEX VAULT)

The Problem:

- ❑ Rising risk of data theft and cyber attacks
- ❑ Sensitive files can be accessed without authorization
- ❑ Existing storage systems often lack strong encryption
- ❑ Users require a secure and easy-to-use protection system

Proposed Solution:

CRYPTEX VAULT

- ❑ Intelligent File Security Platform
- ❑ A web-based application that enables users to securely encrypt and decrypt files using modern cryptographic techniques.

Core Features:

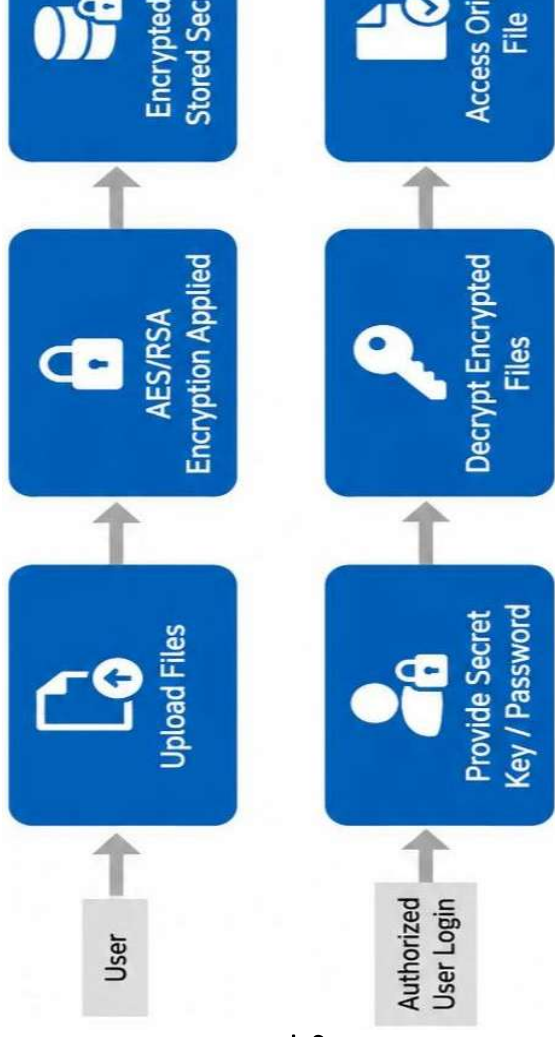
- ❑ AES-based symmetric file encryption/decryption
- ❑ Secure password-based authentication
- ❑ Strong confidentiality and data protection
- ❑ Unauthorized access prevention
- ❑ User-friendly interactive interface
- ❑ Secure cryptographic key management
- ❑ Efficient and scalable security architecture

TECHNICAL APPROACH

Tech stack:

- Frontend: HTML + CSS
- Backend: Python + Flask
- Security: Password-based key derivation using PBKDF2+Salt
- Encryption : Fernet (AES-based encryption)
- Deployment: Render (for hosting the web application)
- Version Control: Git + GitHub

❖ Implementation



FEASIBILITY AND VIABILITY

Technical Feasibility :

- ❑ AES-based encryption implemented using Fernet library
- ❑ Flask enables lightweight and efficient web deployment
- ❑ Supports multiple file formats for encryption/decryption
- ❑ Random salt generation to prevent identical passwords from generating identical encryption keys.

Economic Feasibility :

- ❑ Low development and maintenance cost
- ❑ Open-source technologies reduce expenses
- ❑ Zero licensing cost

Operational Feasibility:

- ❑ Easy for users to operate
- ❑ Minimal training required
- ❑ Simple and user-friendly interface
- ❑ Minimal technical knowledge required for users

Security Feasibility :

- ❑ Password-protected file encryption
- ❑ No permanent storage of files on server

Future Viability :

- ❑ Can be extended with cloud storage
- ❑ Scalable for advanced cybersecurity

IMPACT AND BENEFITS

Cybersecurity Benefits:

- ❑ AES-based encryption protects confidential files from unauthorized access
- ❑ Password-protected decryption ensures secure file recovery
- ❑ Prevents data leakage during online file sharing

User Benefits:

- ❑ Simple and easy-to-use web interface
- ❑ Fast file encryption and decryption process
- ❑ Supports files like PDF, DOCX, ZIP, images, and text files

Technical Benefits:

- ❑ Built using Flask and Fernet (AES-based encryption)
- ❑ In-memory processing avoids permanent storage
- ❑ Lightweight architecture suitable for cloud deployment

Business & Practical Benefits:

- ❑ Low development and deployment costs
- ❑ Useful for students, organizations, and security
- ❑ Scalable for future cybersecurity and cloud applications

RESEARCH AND REFERENCES

Cryptography Research :

- ❑ Studied AES-based encryption and Fernet cryptographic implementation
- ❑ Analyzed password-based encryption and SHA-256 hashing techniques
- ❑ Explored secure file protection methods against unauthorized access

References:

- ❑ Python Cryptography Official Documentation
- ❑ NIST (National Institute of Standards and Technology) AES Standards

Libraries/Docs :

- ❑ Flask Documentation — <https://flask.palletsprojects.com/>
- ❑ Cryptography Library Docs — <https://cryptography.io/en/latest/>
- ❑ Python Documentation — <https://docs.python.org/3/>
- ❑ Render Deployment Docs — <https://render.com/docs>

Project Deployment:

- ❑ Successfully deployed on “Render.com”
- ❑ Publicly accessible web application
- ❑ Supports real-time file encryption and decryption

Website Link:

<https://fileencryption-crypto.onrender.com/>